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# Krylov-space anatomy and spread complexity of a disordered quantum spin chain

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We investigate the anatomy and complexity of quantum states in Krylov space, in the ergodic and many-body localised (MBL) phases of a disordered, interacting spin chain. The Krylov basis generated by the Hamiltonian from an initial state provides a representation in which the spread of the time-evolving state constitutes a basis-optimised measure of complexity. We show that the long-time Krylov spread complexity sharply distinguishes the two phases. In the ergodic phase, the infinite-time complexity scales linearly with the Fock-space dimension, indicating that the state spreads over a finite fraction of the Krylov chain. By contrast, it grows sublinearly in the MBL phase, implying that the long-time state occupies only a vanishing fraction of the chain. Further, the profile of the infinite-time state along the Krylov chain exhibits a stretched-exponential decay in the MBL phase. This behaviour reflects a broad distribution of decay lengthscales, associated with different eigenstates contributing to the long-time state. Consistently, a large-deviation analysis of the statistics of eigenstate spread complexities shows that while the ergodic phase receives contributions from almost all eigenstates, the complexity in the MBL phase is dominated by a vanishing fraction of eigenstates, which have anomalously large complexity relative to the typical ones.

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## I. INTRODUCTION

The many-body localisation transition [1, 2] in disordered quantum many-body systems can be viewed as an eigenstate transition: across a critical disorder strength, the nature of eigenstates at arbitrary finite energy densities changes qualitatively between the ergodic and many-body localised (MBL) phases (see Refs. [3–9] for reviews and further references). One of the most prominent real-space signatures of this transition is encoded in the eigenstate expectation values of local observables. In the ergodic phase, these satisfy the eigenstate thermalisation hypothesis [10–14], whereas in the MBL phase they violate it.

From a purely eigenstate perspective, the distinction between the ergodic and MBL phases is also reflected in bipartite entanglement entropy. Eigenstates in the ergodic phase typically exhibit volume-law entanglement, while those in the MBL phase obey an area law [3, 5, 15–17]. The latter is intimately connected to the existence of an extensive number of (quasi)local integrals of motion [16–18]. Consequently, MBL eigenstates can be related to trivial product states by finite-depth local unitary circuits, whereas no such local circuit exists for ergodic eigenstates [15]. This perspective suggests that eigenstates in the two phases possess fundamentally different complexities, at least in the abstract sense of circuit complexity [19–21]

A complementary and equally insightful approach to the many-body localisation problem exploits the exact mapping of any disordered interacting many-body Hamiltonian to a tight-binding Hamiltonian, describing a fictitious single particle hopping on the associated correlated Fock-space graph of the system; where sites correspond to many-body basis states and links correspond to matrix elements of the Hamiltonian [22–30] (see Refs. [7, 8] for reviews of this approach). The structure of many-

body eigenstates on this graph is qualitatively different in the ergodic and MBL phases, as reflected e.g. in their inverse participation ratios (IPRs). Specifically, MBL eigenstates display multifractal statistics, manifest in anomalous scaling of the IPRs with the Fock-space dimension $N_{\mathcal{H}}$, while ergodic eigenstates are fully extended over the graph [26, 27, 31–34]. Physically, this implies that MBL eigenstates occupy a number of Fock-space sites which, although exponentially large in system size, constitutes a vanishing fraction of the total Fock-space dimension, while ergodic eigenstates by contrast occupy a finite fraction of the graph.

A dynamical manifestation of this distinction arises when the system is initialised in a many-body basis state corresponding to a single site of the Fock-space graph. Under time evolution, the state spreads over the graph. In the ergodic phase, the long-time state occupies a finite fraction of the graph, whereas in the MBL phase it remains confined to a vanishing fraction even at arbitrarily late times [35, 36]. This suggests the existence of a notion of complexity associated with the spread of the state on the Fock-space graph, which should behave qualitatively differently in the two phases.

An intuitive measure of this complexity is the size of the wavefunction on the graph. But such a measure is basis dependent, so this naturally raises the question of whether there exists a basis in which the spread of the state – and hence the associated complexity – is minimised. Importantly, it was shown that the Krylov basis generated by the Hamiltonian from the initial state minimises the complexity of the time-evolving state [37]. Consequently, the spread complexity of the state in Krylov space (hereafter referred to as the Krylov spread complexity) provides a bona fide, basis-optimised measure of complexity.

This in turn motivates the central question of this work: what is the anatomy of states in Krylov space in the ergodic and MBL phases, and how does the Krylov spread complexity distinguish between them? We also show that the Krylov-space anatomy of states provides insights into the MBL phase that are arguably more difficult to obtain from the Fock-space perspective. This is because the Hamiltonian in Krylov space takes the form of a one-dimensional (correlated) disordered tight-binding chain of length $N_{\mathcal{H}}$, where the ordering of basis states and the notion of distance are much simpler than on the high-dimensional Fock-space graphs.

The present work should be contrasted with the much more widely studied operator Krylov complexity [38–51], where the operator growth of an initially real-space-local operator is mapped onto the spread of the operator in the Krylov space of the operator Hilbert space. In fact, the operator Krylov complexity is not necessarily a sensitive probe of the MBL phase, because in this phase too the operator may delocalise in the Krylov space, as shown e.g. for the phenomenological $\ell$-bit model [43] (neither in fact is it a sensitive probe of integrability [52]).

On the other hand, as shown in this work, the Krylov-space anatomy, as well as the Krylov spread complexity of an initial state [53–63] evolved for arbitrarily long (infinite) times, shows qualitative differences in the ergodic and MBL phases. In fact, the Krylov spread complexity has already proven successful in capturing aspects of ergodicity breaking in both single-particle and many-body settings under specific initial conditions [53–59]. More specifically, we find that the infinite-time Krylov spread complexity in the ergodic phase scales linearly with the Fock-space dimension, $\propto N_{\mathcal{H}}$, whereas in the MBL phase it grows sublinearly as $N_{\mathcal{H}}^{\alpha}$ with $\alpha < 1$. Coupled to the fact that the Krylov space is an ordered one-dimensional chain of length $N_{\mathcal{H}}$, the above scaling presents a simple yet insightful picture: in the ergodic phase the infinite-time states extend over a finite fraction of the entire Krylov chain, while in the MBL phase, they extend to only a vanishing fraction of the chain.

In the MBL phase we find in fact that the profile of the infinite-time state along the length of the Krylov chain has a non-trivial decay described by a stretched exponential. We present a phenomenological theory which shows how such a profile can emerge, from the interplay of exponential decay of the state amplitude on the Krylov chain and a broad (but not itself heavy-tailed) distribution of associated decay length scales. Importantly, these distributions arise not only over disorder realisations, but also across eigenstates within a given realisation. This interplay becomes evident when the infinite-time Krylov spread complexity is expressed as a sum over eigenstates of the Krylov spread complexities associated with individual eigenstates.

The latter decomposition yields further insight into the nature of eigenstates in the two phases. In the ergodic phase, eigenstate spread complexities are found to have a narrow distribution over the eigenstates. By contrast, in the MBL phase the distribution is broad, and its tails become heavier with increasing system size. Using a large-deviation analysis, we show that in the ergodic phase the infinite-time Krylov spread complexity receives contributions from almost all eigenstates, whereas in the MBL phase it is dominated by only a vanishing fraction of eigenstates – albeit exponentially many in system size – lying in the tails of the distribution. This provides a manifestation, on the Krylov chain, of rare resonances characteristic of the MBL phase [28, 64–74].

The paper is organised as follows. Sec. II reviews briefly the definition of the Krylov spread complexity and the construction of the Krylov basis. In Sec. III, we specify the disordered, tilted-field Ising chain as our model of choice, and discuss its Krylov chain and its conceptual connections to the model's Fock-space graph. Detailed numerical results for the Krylov-space anatomy and the spread complexity, along with a phenomenological theory in the MBL phase, constitute Sec. IV. The large-deviation analysis of the statistics of the eigenstate spread complexities contributing to the infinite-time spread complexity is presented in Sec. V. Some concluding remarks are given in Sec. VI.

## II. KRYLOV SPREAD COMPLEXITY

We start by discussing the construction of the Krylov chain, and formally defining the associated spread complexity [37]. This enables us to discuss several important conceptual points, and to put on a concrete footing the motivation behind studying the anatomy of states on the Krylov chain.

Consider a general many-body Hamiltonian of form

$$H = H_0 + H_1 \,, \tag{1}$$

denoting by $\{|I\rangle\}$ the eigenstates of $H_0$, and by $\{|E\rangle\}$ the eigenstates of $H$ with eigenvalue $E$. The setting in which we will be interested is initialising the state of system at $t = 0$ in an eigenstate of $H_0$, $|\psi_{t=0}\rangle = |I\rangle$, and asking how *complex* the state becomes upon evolution over an arbitrarily long time under the full Hamiltonian $|\psi_t\rangle = e^{-iHt}|\psi_0\rangle$. Denoting by $S_K(t)$ the measure of complexity for the state at time $t$, our specific focus will be the long-time limit $S_{K,\infty} \equiv \lim_{t\to\infty} S_K(t)$. This can be expressed in terms of correlated eigenstate amplitudes on the Krylov chain, as we show below.

For a general state $|\psi_t\rangle$, given some orthonormal basis $\{|\mathcal{V}_n\rangle\}$, this can be quantified using a cost function

$$C_{\mathcal{V}} = \sum_n \mu_n |\langle\psi_t|\mathcal{V}_n\rangle|^2 \,, \tag{2}$$

where the $\mu_n$s are a positive, increasing sequence of real numbers. It is customary [37] to consider $\mu_n = n$, such that $C_{\mathcal{V}}$ can be interpreted as the size of the support of the state $|\psi_t\rangle$ assuming the basis $\{|\mathcal{V}_n\rangle\}$ to be ordered. A natural measure of complexity is therefore to consider the basis which minimises the cost function. An important result in this context is that the basis which minimises Eq. 2 is the Krylov basis [37], generated via the Hamiltonian $H$, with its first basis vector given by $|\psi_0\rangle$. This is an ordered orthonormal basis, the basis states being denoted by $|k_n\rangle$ with $n = 0, 1, \cdots, N_{\mathcal{H}} - 1$, where $N_{\mathcal{H}}$ is the Fock-space dimension of the system. These Krylov basis states are defined via the recursion relation [75]

$$|k_n\rangle = \frac{1}{b_n}\left[H|k_{n-1}\rangle - a_{n-1}|k_{n-1}\rangle - b^*_{n-1}|k_{n-2}\rangle\right] , \tag{3}$$

where $a_n = \langle k_n|H|k_n\rangle$ and $b_n = \langle k_n|H|k_{n-1}\rangle$, and $|k_0\rangle = |I\rangle$ is the boundary condition for the recursion.

An immediate consequence of this construction is that the many-body Hamiltonian in the Krylov basis takes the tridiagonal form

$$H = \sum_{n=0}^{N_{\mathcal{H}}-1} a_n |k_n\rangle\langle k_n| + \sum_{n=1}^{N_{\mathcal{H}}-1} \left[b_n |k_{n-1}\rangle\langle k_n| + \text{h.c.}\right] . \tag{4}$$

The Krylov basis is thus equivalent to a one-dimensional, nearest-neighbour, tight-binding chain of length $N_{\mathcal{H}}$, described by the Hamiltonian Eq. 4. While the fact that the length of the Krylov chain is $N_{\mathcal{H}}$ is a straightforward fallout, its conceptual importance is that it provides us with a naturally ordered one-dimensional basis for the many-body Hilbert space, which has a simple and unambiguous notion of distance.

With the initial state $|I\rangle = |k_0\rangle$ localised at one end of the chain, the temporal evolution of the state on the Krylov chain is encoded in the probability amplitudes

$$c_n(t) = \langle k_n|\psi_t\rangle = \sum_E e^{-iEt}\langle k_n|E\rangle\langle E|k_0\rangle \,. \tag{5}$$

The spread complexity is then defined as [37]

$$S_K(t) = \sum_{n=0}^{N_{\mathcal{H}}-1} n|c_n(t)|^2 \,, \tag{6}$$

and the one-dimensional nature of the Krylov chain leads to the interpretation of the spread complexity as simply the size of the support of the wavefunction on the chain.

We will be particularly interested in the spread complexity at infinite time, formally defined as

$$S_{K,\infty} = \lim_{t\to\infty} \frac{1}{t}\int_0^t dt' \; S_K(t') \,. \tag{7}$$

This can be conveniently expressed as

$$S_{K,\infty} = \sum_{n=0}^{N_{\mathcal{H}}-1} n\Lambda_n \,, \tag{8}$$

where (with $|E\rangle$ denoting an eigenstate of $H$)

$$\Lambda_n = \sum_E \Lambda_{n,|E\rangle} \text{ with } \Lambda_{n,|E\rangle} = |\langle k_n|E\rangle\langle E|k_0\rangle|^2 \,. \tag{9}$$

Equations 8 and 9 relate the infinite-time spread complexity to eigenstate amplitudes on the Krylov chain. Note that conservation of probability trivially leads to $\Lambda_n$ satisfying the sum rule

$$\sum_{n=0}^{N_{\mathcal{H}}-1} \Lambda_n = 1 \,. \tag{10}$$

As such, $\Lambda_n$ can interpreted as a probability distribution over the Krylov chain, for which $S_{K,\infty}$ is nothing but its first moment. The profile of $\Lambda_n$ on the Krylov chain, and its scaling with $N_{\mathcal{H}}$, together with the statistical properties of the eigenstate spread complexity $S_{K,\infty}$, will be the central quantities of interest in this work.

## III. DISORDERED SPIN-1/2 CHAIN AND ITS KRYLOV CHAIN

### A. Model

As a concrete setting for our studies, we consider the disordered tilted-field Ising (TFI) spin-1/2 chain, which

has emerged as the standard archetype for a disordered system hosting a MBL phase at strong disorder [8, 9, 76, 77]. The model is described by the Hamiltonian

$$H = \underbrace{\sum_{i=1}^{L-1} J_i\, \sigma_i^z \sigma_{i+1}^z + \sum_{i=1}^{L} h_i \sigma_i^z}_{H_0} + \underbrace{\Gamma \sum_{i=1}^{L} \sigma_i^x}_{H_1}, \tag{11}$$

where (taking $\Gamma \equiv 1$) $J_i \in [0.8, 1.2]$ and $h_i \in [-W, W]$ are independent random numbers. This model hosts an MBL phase for strong enough disorder, and an ergodic phase at weaker disorder strengths. For the parameters above, within accessible system sizes, the critical disorder strength is $W \simeq 3.7$ [77].

Given that $H_0$ in Eq. 11 is composed solely of $\{\sigma_i^z\}$ operators, the basis $\{|I\rangle\} = \{|\{\sigma_i^z\}\rangle\}$ is simply the $\sigma^z$-product state basis. The $0^{\text{th}}$ Krylov orbital, $|k_0\rangle$, is then simply a specific $\sigma^z$-product state, which we denote by $|I_0\rangle$. In our numerical calculations we consider $|I_0\rangle$ to be the $\sigma^z$-product state whose energy is closest to the middle of the many-body spectrum. With this choice, the entire set of Krylov-basis vectors $\{|k_n\rangle\}$ together with the parameters $\{a_n\}$ and $\{b_n\}$ describing the Krylov Hamiltonian, can be computed explicitly. Some details of the statistical properties of the $\{a_n, b_n\}$ are relegated to Appendix A, but one essential point should be mentioned here. Specifying an instance of the Hamiltonian in Eq. 11 requires only $\propto L$ independent random values of $J_i$ and $h_i$. By contrast, specifying the Krylov Hamiltonian requires $\propto N_{\mathcal{H}}$ values of $a_n$ and $b_n$ (Eq. 4). However, since the latter $\propto N_{\mathcal{H}}$ coefficients are constructed from only $\propto L$ independent $J_i$ and $h_i$ values, strong correlations between the matrix elements of the Krylov Hamiltonian necessarily arise. As a result, the problem is very different from a conventional Anderson localisation problem [78–80], despite the Krylov chain being one dimensional.

### B. On the connection between Krylov and Fock-space bases

Before delving into the anatomy of eigenstates on the Krylov chain, it is important from a conceptual point of view to understand the connection between the Krylov basis and the more commonly employed Fock-space basis, and its associated Fock-space graph.

The eigenstates of $H_0$ ($\sigma^z$-product states $\{|I\rangle\}$) form a natural choice of basis for the Fock space. Properties of the Fock-space graph have been discussed in detail previously (see [8] and references therein); here we summarise the features of it essential for the present work.

The Fock-space graph for the TFI chain, illustrated as part of Fig. 1, is an $L$-dimensional hypercube with $N_{\mathcal{H}} = 2^L$ nodes, each such representing a $\sigma^z$ configuration. Each node has a connectivity of precisely $L$, which corresponds to flipping each of the $L$ spins on the chain

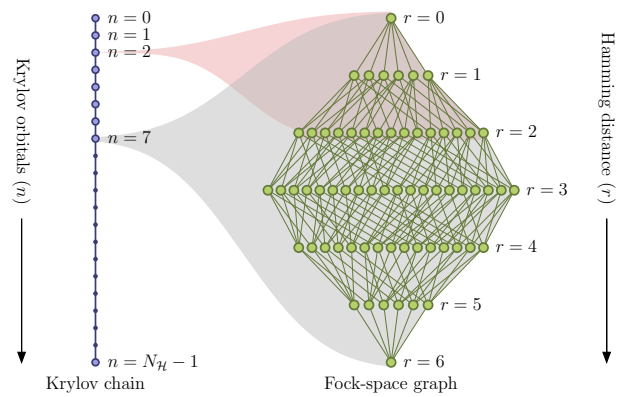


FIG. 1. In the Krylov-space basis, the many-body Hamiltonian takes the form of a nearest-neighbour tight-binding chain (Eq. 4), of length $N_{\mathcal{H}} = 2^L$, with the orbitals $|k_n\rangle$ labelled $n = 0, 1, \cdots, N_{\mathcal{H}} - 1$. On the Fock-space graph (illustrated for $L = 6$), the Krylov $|k_0\rangle$ orbital is identified with the top apical site $|I_0\rangle$. For $n \leq L$, $|k_n\rangle$ is supported on Fock-space sites which lie within Hamming distance $r \leq n$, as illustrated by the red-shaded connection between $|k_2\rangle$ and all Fock-space sites with $r \leq 2$. By contrast, for $L + 1 \leq n \leq N_{\mathcal{H}} - 1$, $|k_n\rangle$ is in general supported on the entire Fock-space graph, as indicated by the grey-shaded connections.

under the action of $H_1$ ($= \Gamma \sum_{i=1}^{L} \sigma_i^x$). The Hamming distance between two $\sigma^z$ configurations – defined as the number of sites where the two configurations differ from each other – endows the Fock-space graph with a natural notion of distance. For the form of $H_1$ in Eq. 11 it is also, conveniently, the shortest path between the two nodes. The above features imply that for a given $|I_0\rangle$, the remaining nodes can be arranged in a row-wise fashion such that all nodes on row $r$ are a Hamming distance of $r$ from $|I_0\rangle$. Naturally there are $L$ such rows, with the $r^{\text{th}}$ row containing $N_r = \binom{L}{r}$ nodes. The form of $H_1$ in Eq. 11 also implies that any node on row $r$ is connected to nodes only on rows $r \pm 1$.

Given the structure of the Fock-space graph, a natural question is: what is the distribution of the Krylov-basis vectors, $\{|k_n\rangle\}$, on the Fock-space graph? To make this quantitative, define the overlap of the Krylov-basis states and the Fock-space basis states as

$$w_{nI} = |\langle k_n | I \rangle|^2, \tag{12}$$

and the average weight of the $n^{\text{th}}$ Krylov-basis state at Hamming distance $r$ from $I_0$ on the Fock-space graph as

$$w_n(r) = \frac{1}{N_r} \sum_{I: r_{II_0} = r} w_{nI} = \frac{1}{\binom{L}{r}} \sum_{I: r_{II_0} = r} |\langle k_n | I \rangle|^2. \tag{13}$$

Considering $|k_0\rangle = |I_0\rangle$, the construction of the Krylov chain in Eq. 3, together with the form of $H_1$, makes it clear that for $n \leq L$, the Krylov $|k_n\rangle$ has support on Fock-space sites which lie at Hamming distances $r \leq n$ from $I_0$. However, for $n > L$, the orthonormality of $|k_n\rangle$ with

$|k_m\rangle$ $\forall\, m < n$ implies that the support of $|k_n\rangle$ folds back on the Fock-space graph, and thus in principle exists on the entire graph. This is shown schematically in Fig. 1 for the case of $L = 6$. Indeed, since the Krylov chain has a length $N_{\mathcal{H}} = 2^L$, a fraction tending to unity of the sites thereon potentially have support over the entire Fock space.

It is of course important to establish that the Krylov-basis states indeed span the Fock-space graph, whether in the ergodic or MBL regimes. To this end, we define an effective size of the wave function corresponding to the Krylov orbital $|k_n\rangle$ on the Fock-space graph, via

$$\mathcal{R}_n = \sum_{r=0}^{L} r \binom{L}{r} w_n(r) = \sum_{r=0}^{L} r \sum_{I: r_{I I_0} = r} w_{nI} \,. \tag{14}$$

Physically, $\mathcal{R}_n$ gives the mean Hamming distance on the Fock-space graph that is associated with the given Krylov $|k_n\rangle$, and as such is a bona fide measure of how the Krylov-basis state is spread over the Fock-space graph. Since $\{|k_n\rangle\}$ forms a complete orthonormal basis, such that $\sum_n w_n(r) = 1$ (Eq. 13), note that $\mathcal{R}_n$ satisfies a sum rule

$$\sum_{n=0}^{N_{\mathcal{H}}-1} \mathcal{R}_n = \sum_{r=0}^{L} r \binom{L}{r} = N_{\mathcal{H}} L/2 \,. \tag{15}$$

Sufficiently deep in the ergodic phase, one expects $w_{nI} \sim N_{\mathcal{H}}^{-1}$ and hence $w_n(r) \sim N_{\mathcal{H}}^{-1}$ for all $I$ and $n > L$; which when used in Eq. 14 yields $\mathcal{R}_n \simeq L/2$. That this is indeed the case is confirmed by numerical results shown in Fig. 2(a) for a disorder strength $W = 1$ deep in the ergodic phase (and with system sizes $L$ as indicated).

The natural generalisation of this to encompass the MBL regime is the binomial form

$$w_n(r) = p_n^r (1 - p_n)^{L-r} \,, \tag{16}$$

which from Eq. 14 yields $\mathcal{R}_n = p_n L$ (with the deep ergodic phase corresponding to $p_n = 1/2$). The sum rule in Eq. 15 then implies $\sum_n p_n = N_{\mathcal{H}}/2$. Assuming $p_n = g(n/N_{\mathcal{H}}^a)$ to be some function of $n/N_{\mathcal{H}}^a$ (with exponent $a$), this sum rule can be cast for $N_{\mathcal{H}} \gg 1$ as

$$\int_0^{N_{\mathcal{H}}} dn\; g\left(\frac{n}{N_{\mathcal{H}}^a}\right) = N_{\mathcal{H}}^a \int_0^{N_{\mathcal{H}}^{1-a}} dx\; g(x) = \frac{N_{\mathcal{H}}}{2} \,, \tag{17}$$

which naturally requires $a = 1$. This then implies that $\mathcal{R}_n$ satisfies the scaling form

$$\mathcal{R}_n = L \times g\left(\frac{n}{N_{\mathcal{H}}}\right) \,, \tag{18}$$

such that $\mathcal{R}_n/L$ is a function solely of $n/N_{\mathcal{H}}$. While the scaling form is quite trivially satisfied in the ergodic regime (as above), numerical results in the MBL phase, shown in Fig. 2(b) for disorder strength $W = 10$, confirm this to be case in general. The key upshot of this scaling form is that in both the ergodic and MBL regimes, the Krylov-basis states indeed have access to the entire Fock-space graph [1].

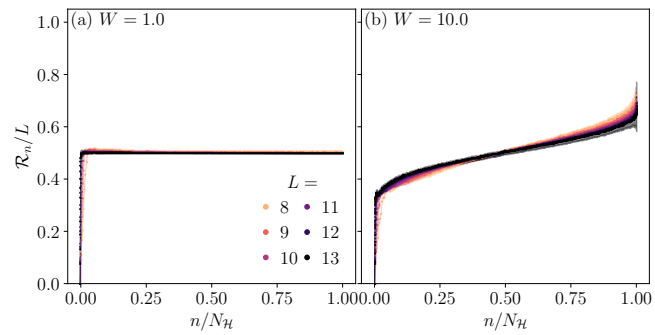


FIG. 2. Numerical results for the effective size of the wave function on the Fock-space graph corresponding to the $n^{\text{th}}$ Krylov orbital, as quantified by $\mathcal{R}_n$ defined in Eq. 14 (with system sizes $L$ indicated). Panels (a) and (b) with $W = 1$ and $W = 10$ exemplify, respectively, the ergodic and MBL regimes. The results conform to the scaling form in Eq. 18. The (thin) shaded region around the data points reflects the statistical errors over disorder realisations.

## IV. SCALING FORMS FOR SPREAD COMPLEXITY AND EIGENSTATE AMPLITUDES

With the basic properties of the Krylov chain and the spread complexity defined, we turn now to the Krylov-space anatomy of the eigenstates.

### A. Distribution and scaling of $S_{K,\infty}$

We start with the distribution of the infinite-time spread complexity over disorder realisations; in particular the distribution $P_s(s)$ of the late-time complexity rescaled by its mean, $s = S_{K,\infty}/\langle S_{K,\infty}\rangle$, numerical results for which are shown in Fig. 3 for representative $W$'s in each of the ergodic and MBL phases.

In the ergodic regime the distributions are well described by a Gaussian with mean 1

$$P_s(s) = \frac{1}{\sqrt{2\pi\sigma_s^2}} \exp\left[-\frac{(s-1)^2}{2\sigma_s^2}\right] \,, \tag{19}$$

and clearly narrow rapidly with increasing system size. Results for the mean $\langle S_{K,\infty}\rangle$ are given in Fig. 4(a), from which it is evident that in the ergodic regime,

$$\langle S_{K,\infty}\rangle \simeq N_{\mathcal{H}}/2 \,. \tag{20}$$

[1] This is not of course the case for the first $L$ Krylov basis states, but that is immaterial since these constitute an exponentially small (in $L$) fraction of the $2^L$ Krylov states.

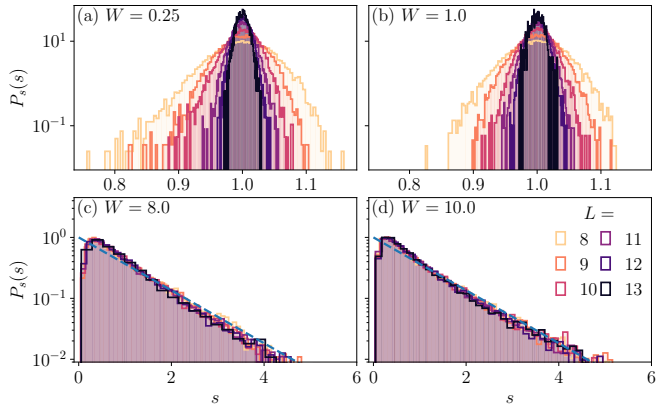


FIG. 3. Distribution $P_s(s)$ of the infinite-time spread complexity rescaled by its mean, $s \equiv S_{K,\infty}/\langle S_{K,\infty}\rangle$, over disorder realisations in both the ergodic [panels (a), (b)] and MBL [panels (c), (d)] regimes, for different $L$. In the ergodic phase the distribution clearly approaches a Gaussian, whereas in the MBL phase it is an exponential distribution, as indicated by the blue dashed lines denoting $e^{-s}$.

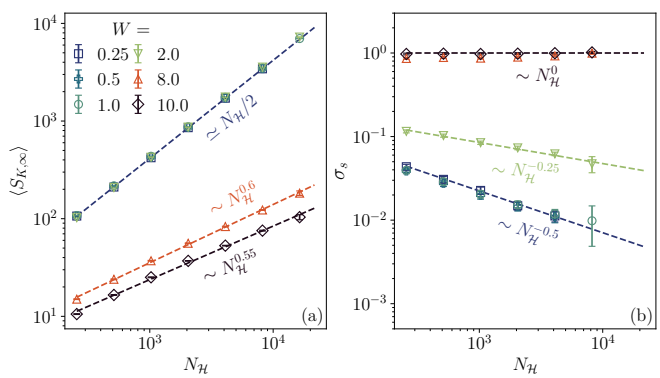


FIG. 4. (a) Disorder averaged $\langle S_{K,\infty}\rangle$ as a function of $N_{\mathcal{H}}$ on logarithmic scales, for different disorder strengths $W$. The straight line fits show that $\langle S_{K,\infty}\rangle \sim N_{\mathcal{H}}^{\alpha}$ with $\alpha = 1$ in the ergodic phase and $\alpha < 1$ in the MBL phase, as indicated explicitly next to fits. (b) Scaling with $N_{\mathcal{H}}$ of the standard deviation of $S_{K,\infty}$ relative to its mean, $\sigma_s \equiv \mathrm{std}[S_{K,\infty}]/\langle S_{K,\infty}\rangle$. The scaling exponents $\varsigma$, defined via $\sigma_s \sim N_{\mathcal{H}}^{-\varsigma}$, are mentioned explicitly next to the fits. In the deep ergodic regime $\varsigma = 1/2$, whereas in the MBL regime $\varsigma \simeq 0$.

Corresponding results for the standard deviation $\sigma_s$, given in Fig. 4(b), show that it decays exponentially with system size as

$$\sigma_s = \frac{\mathrm{std}[S_{K,\infty}]}{\langle S_{K,\infty}\rangle} \sim N_{\mathcal{H}}^{-\varsigma}\,, \tag{21}$$

where $0 < \varsigma < 1$ decreases with increasing $W$. In particular, note that deep in the ergodic phase $\varsigma = 1/2$. Since $\sigma_s$ is exponentially small in the system size $L$, $\lim_{L\to\infty} P_s(s) \to \delta(s-1)$ is thus $\delta$-distributed in the thermodynamic limit. Recalling that $S_{K,\infty}$ is a measure of the size of the late-time wave function on the Krylov chain, the above results imply that in the ergodic regime the wave function is maximally spread out over the entire Krylov chain, with negligible fluctuations in the spread.

The observations above can be understood by considering the eigenvectors deep in the ergodic phase to be well represented by Gaussian random vectors. As such one considers the $\langle k_n|E\rangle \sim \mathcal{N}(0, N_{\mathcal{H}}^{-1/2})$ to be uncorrelated Gaussian random numbers with zero mean and standard deviation $N_{\mathcal{H}}^{-1/2}$; which implies that $\Lambda_{n,|E\rangle} = |\langle k_0|E\rangle|^2|\langle k_n|E\rangle|^2$ (Eq. 9) is a random number with mean $N_{\mathcal{H}}^{-2}$ and standard deviation $2\sqrt{2}N_{\mathcal{H}}^{-2}$ (for any $n > 0$). Using this, it is easily shown that $\Lambda_n = \sum_n \Lambda_{n,|E\rangle}$ is a random number with mean $N_{\mathcal{H}}^{-1}$ and standard deviation $2\sqrt{2}N_{\mathcal{H}}^{-3/2}$. So from Eq. 8,

$$S_{K,\infty} = \sum_{n=0}^{N_{\mathcal{H}}-1} Z_n \tag{22}$$

where $Z_n$ $(= n\Lambda_n)$ is a random number with mean $nN_{\mathcal{H}}^{-1}$ and standard deviation $2\sqrt{2}nN_{\mathcal{H}}^{-3/2}$. The $Z_n$'s are not however independent, as is explicit from the fact that $\langle Z_n Z_m\rangle \neq \langle Z_n\rangle\langle Z_m\rangle$ for $n \neq m$. More specifically, under the assumption above that $\langle k_n|E\rangle \sim \mathcal{N}(0, N_{\mathcal{H}}^{-1/2})$, it can be shown that for $n \neq m$

$$\langle Z_n Z_m\rangle - \langle Z_n\rangle\langle Z_m\rangle = 2nm/N_{\mathcal{H}}^3\,. \tag{23}$$

The covariance matrix for the random variables $\{Z_n\}$ therefore has the form

$$[C_Z]_{nm} = \delta_{nm}\frac{8n^2}{N_{\mathcal{H}}^3} + (1-\delta_{nm})\frac{2nm}{N_{\mathcal{H}}^3}\,. \tag{24}$$

Given that $S_{K,\infty}$ is a sum of $N_{\mathcal{H}}$ correlated random numbers, it follows that for large $N_{\mathcal{H}}$ it is Gaussian distributed (invoking a multivariate Lyapunov version of the central limit theorem); as indeed is corroborated by the numerical results in Fig. 3[(a) and (b)]. The mean of this distribution is given by

$$\langle S_{K,\infty}\rangle = \sum_{n=0}^{N_{\mathcal{H}}-1}\langle Z_n\rangle = \frac{N_{\mathcal{H}}-1}{2} \overset{N_{\mathcal{H}}\gg 1}{\sim} \frac{N_{\mathcal{H}}}{2}\,, \tag{25}$$

as also found numerically in Fig. 4(a). The corresponding variance is given by

$$\mathrm{var}[S_{K,\infty}] = \sum_{n,m}[C_Z]_{nm} = \frac{(N_{\mathcal{H}}^2+3N_{\mathcal{H}}-2)(N_{\mathcal{H}}-1)}{2N_{\mathcal{H}}^2}$$
$$\Rightarrow \mathrm{var}[S_{K,\infty}] \overset{N_{\mathcal{H}}\gg 1}{\sim} \frac{N_{\mathcal{H}}}{2}\,. \tag{26}$$

This implies that in the ergodic regime, $\sigma_s \sim N_{\mathcal{H}}^{-1/2}$ which is indeed exactly the result found numerically in Fig 4(b) for sufficiently weak disorder. It is worth reiterating the conceptual point that, even though the eigenvector amplitudes on the Krylov chain in the weak-disorder ergodic regime can be considered to be uncorrelated random variables, the $\Lambda_n$'s (or equivalently the

$Z_n = n\Lambda_n$) are in fact correlated for different $n$. This correlation is rooted in the presence of the common factors of $|\langle E|k_0\rangle|^2$in both $\Lambda_n$ and $\Lambda_m$ (see Eq. 9) even for $n \neq m$.

The physical content of Eq. 25 is also quite transparent. Recall that $\langle S_{K,\infty}\rangle$ gives the mean position on the Krylov chain, as $t \to \infty$, following initiation of the system at the end of the chain (in $|k_0\rangle$). In the ergodic phase, with eigenstates delocalised over the chain, the long-time mean position should thus be the midpoint of the chain (the length of which is $N_{\mathcal{H}}$); i.e. $\langle S_{K,\infty}\rangle = N_{\mathcal{H}}/2$, as in Eq. 25.

The situation in the MBL regime is completely different. The distribution of $S_{K,\infty}$, or equivalently $P_s$, is qualitatively different from that in the ergodic regime. Results for the mean $\langle S_{K,\infty}\rangle$ in Fig. 4(a) show that

$$\langle S_{K,\infty}\rangle \sim N_{\mathcal{H}}^{\alpha} \quad : \ \alpha < 1 \tag{27}$$

with an exponent $\alpha < 1$, which decreases with increasing disorder strength. The ratio $\langle S_{K,\infty}\rangle / N_{\mathcal{H}} \propto N_{\mathcal{H}}^{\alpha-1}$ gives the fraction of the Krylov-chain length which is accessed in the long-time limit following initiation of the system in $|k_0\rangle$. Since $\alpha < 1$, it thus vanishes exponentially in system size $L$, which is of course symptomatic of the many-body localised character of the eigenstates.

In marked contrast to its Gaussian counterpart in the ergodic phase, the distribution of $s = S_{K,\infty}/\langle S_{K,\infty}\rangle$ is very well described by an exponential form

$$P_s(s) = e^{-s}\,, \tag{28}$$

as seen in Fig. 3[(c), (d)]. This is significant, as it implies that the standard deviation of $S_{K,\infty}$ scales in the same way with $N_{\mathcal{H}}$ as the mean,

$$\mathrm{std}[S_{K,\infty}] \sim N_{\mathcal{H}}^{\alpha}\,. \tag{29}$$

The net physical import of the above results is the following. In the ergodic regime $S_{K,\infty}$ is sharply distributed (becoming a $\delta$-function as $L \to \infty$) around its mean, which scales $\sim N_{\mathcal{H}}$, reflecting the ergodic and extended nature of the eigenstates on the Krylov chain. By contrast, in the MBL regime $S_{K,\infty} \sim N_{\mathcal{H}}^{\alpha}$ with $\alpha < 1$, reflecting the non-ergodic nature of the eigenstates. In addition, the distribution of $S_{K,\infty}$ is rather broad, as the mean and standard deviation scale in the same way with $N_{\mathcal{H}}$; as we will see in Sec. V, this is related to a large-deviation character of the contributions to $S_{K,\infty}$ from different eigenstates.

### B. Profile of $\Lambda_n$ on Krylov chain

The quantity $\Lambda_n$ (Eq. 9) gives the long-time probability $|c_n(t\to\infty)|^2$ (Eqs. 5,6) that the system will be found on orbital $n$ of the Krylov chain, given its initiation in the $n = 0$ orbital. While $S_{K,\infty} = \sum_n n\Lambda_n$ is itself the first moment of this distribution, the profile of the wave function, encoded in $\Lambda_n$, clearly contains finer information about the anatomy of the state on the Krylov chain. It is to this that we now turn.

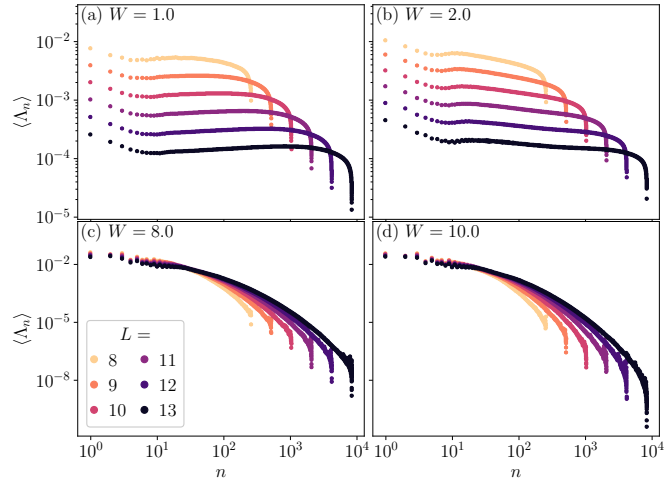


FIG. 5. Profile of the disorder-averaged $\langle\Lambda_n\rangle$ (defined in Eq. 9) on the Krylov chain, for different $L$. Panels (a) and (b) correspond to the ergodic phase for $W = 1, 2$, while panels (c) and (d) correspond to the MBL phase for $W = 8, 10$.

Fig. 5 shows results for the disorder-averaged $\langle\Lambda_n\rangle$ *vs* $n$ ($\in [0, N_{\mathcal{H}} - 1]$), in both the ergodic regime (panels (a)-(b)) as well as the MBL regime (panels (c)-(d)). In the ergodic regime, following a 'transient' behaviour for $n \leq L$, the profile is approximately a flat plateau with $n$, indicating the ergodicity of all eigenstates. Of course, the normalisation $\sum_{n=0}^{N_{\mathcal{H}}-1}\Lambda_n = 1$ implies that the height of this plateau decreases exponentially with $L$, since its support itself grows exponentially with $L$. In the MBL regime on the other hand, there is a systematic decay of $\langle\Lambda_n\rangle$ with $n$, but the decay is slower for larger $L$ which suggests that $\langle\Lambda_n\rangle$ decays with $n$ over scales which scale as $N_{\mathcal{H}}^{\beta}$. This motivates the general scaling ansatz

$$\langle\Lambda_n\rangle = N_{\mathcal{H}}^{-\beta} f\left(\frac{n}{N_{\mathcal{H}}^{\beta}}\right)\,, \tag{30}$$

with $\beta \leq 1$ and the equality expected to be satisfied in the ergodic phase. For $N_{\mathcal{H}} = 2^L \gg 1$, this implies for $\langle S_{K,\infty}\rangle = \sum_n n\,\langle\Lambda_n\rangle$

$$\begin{aligned}\langle S_{K,\infty}\rangle &= N_{\mathcal{H}}^{-\beta}\int_0^{N_{\mathcal{H}}} dn\; n\; f\left(\frac{n}{N_{\mathcal{H}}^{\beta}}\right)\\ &\simeq N_{\mathcal{H}}^{\beta}\int_0^{\infty} dx\; x f(x)\,.\end{aligned} \tag{31}$$

However the results in Eq. 20 and Eq. 27 for the scaling of $\langle S_{K,\infty}\rangle$ with $N_{\mathcal{H}}$ imply that $\beta = \alpha$. Indeed, numerical results presented in Fig. 6 show that the data for $\langle\Lambda_n\rangle$ is in excellent agreement with the scaling ansatz in Eq. 30 with $\beta = \alpha$. This shows how the profile of the decay of

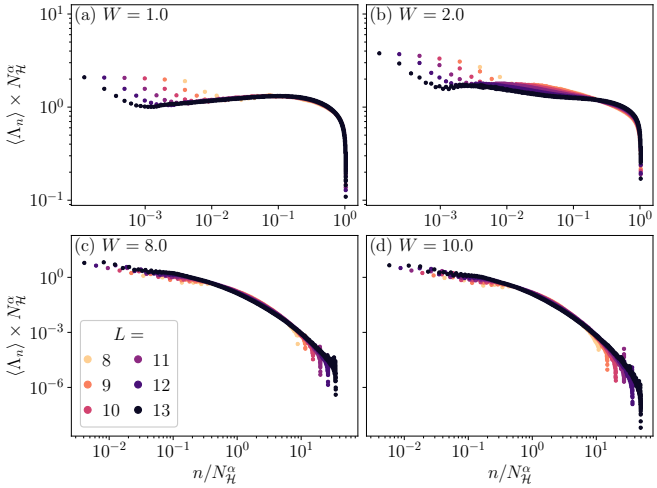


FIG. 6. Scaling of the $\langle \Lambda_n \rangle$ profile. $\langle \Lambda_n \rangle \times N_{\mathcal{H}}^{\alpha}$ is plotted *vs* $n/N_{\mathcal{H}}^{\alpha}$ ($\equiv x$), with the exponents $\alpha$ obtained from the scaling of $\langle S_{K,\infty} \rangle$ in Fig. 4. The data demonstrates the scaling form in Eq. 30 (with $\beta = \alpha$). Panels (a) and (b) correspond to the ergodic phase for $W = 1, 2$, while panels (c) and (d) correspond to the MBL phase for $W = 8, 10$.

$\langle \Lambda_n \rangle$ with $n$, and its dependence on $N_{\mathcal{H}}$, is intimately connected to the scaling of $\langle S_{K,\infty} \rangle$ with $N_{\mathcal{H}}$.

We now consider the functional form of the scaling function, $f(x)$, in Eq. 30. In the ergodic phase, as discussed above, $f(x) \approx 1$ is an approximately flat function. On the other hand, as shown in panels (c) and (d) of Fig. 6, in the MBL phase $f(x)$ decays systematically with $n$. Note that on logarithmic axes, the latter data curves downwards, indicating a decay faster than any power law. This leads us to posit that the decay is exponential, possibly stretched. To confirm this, in Fig. 7 we plot $-\ln[f(x)]$ as a function of $x$, on logarithmic axes. The results are consistent with $-\ln[f(x)] \sim x^{1/2}$, which in turn implies a stretched exponential decay of $\Lambda_n$,

$$\langle \Lambda_n \rangle \sim N_{\mathcal{H}}^{-\alpha} \exp\left[-c\left(\frac{n}{N_{\mathcal{H}}^{\alpha}}\right)^{\gamma}\right], \tag{32}$$

where $\gamma \simeq 1/2$ (and $c > 0$). While $\gamma \simeq 1/2$ is a good description of the stretched exponential for the bulk of the data, we note that for the larger values of $n/N_{\mathcal{H}}^{\alpha}$, the data is better described by $\gamma = 1/3$ (grey dashed lines in Fig. 7); we will return to this possibility shortly. While the precise value of the stretch exponent is slightly ambiguous within our numerical results, what stands robustly is the qualitative statement that $\langle \Lambda_n \rangle$ in the MBL regime has a stretched-exponential profile.

### C. Phenomenological theory

In this section, we provide a simple phenomenological theory for the numerical results presented above, specifically in the MBL regime. In particular, this provides

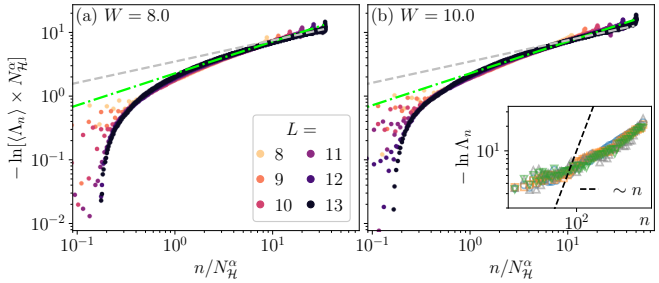


FIG. 7. Stretched-exponential behaviour of the scaling function for $\langle \Lambda_n \rangle$ (see Eq. 30) with $\beta = \alpha$ in the MBL phase (and the $\alpha$ values those obtained in Fig. 4). Plots show the negative of the logarithm of the scaling function on logarithmic axes, where a straight line with slope $< 1$ denotes a stretched exponential. The green dashed-dotted and the grey dashed lines correspond respectively to a stretch exponent of $\gamma = 1/2$ and $1/3$ (see Eq. 32). Inset in (b) shows data for a few individual disorder realisations (different colours), each of which are also unambiguously stretched exponentials as the slope is clearly $< 1$; the black dashed line denotes a slope equal to 1.

a plausible rationalisation of the stretched exponential decay of $\langle \Lambda_n \rangle$ with $n$ as well as the exponential distribution of $S_{K,\infty}$ over disorder realisations. The two key ingredients that enter the phenomenological picture are:

(i) For a given disorder realisation, there is a normalised distribution over eigenstates, $P_{\xi_{|E\rangle}}(\xi)$, of lengthscales $\xi_{|E\rangle}$ on the Krylov chain, of form

$$P_{\xi_{|E\rangle}}(\xi) = \frac{1}{\xi_{\rm d}} \times p_{\xi_{|E\rangle}}(\xi/\xi_{\rm d}) \,. \tag{33}$$

The distribution has a characteristic lengthscale $\xi_{\rm d}$, which without loss of generality can be defined via its first moment, $\xi_{\rm d} = \int d\xi \; \xi \; P_{\xi_{|E\rangle}}(\xi)$.

(ii) Over an ensemble of disorder realisations, $\xi_{\rm d}$ itself has a distribution $P_{\xi_{\rm d}}(\xi_{\rm d})$ of form

$$P_{\xi_{\rm d}}(\xi_{\rm d}) = \frac{1}{\bar{\xi}} \times p_{\xi_{\rm d}}(\xi_{\rm d}/\bar{\xi}), \tag{34}$$

and is likewise characterised by a lengthscale defined by $\bar{\xi} = \int d\xi_{\rm d} \; \xi_{\rm d} \; P_{\xi_{\rm d}}(\xi_{\rm d})$.

The starting point of the phenomenological picture is the conjecture that, for a given disorder realisation,

$$\begin{aligned} \Lambda_n &= \sum_E |\langle k_0|E\rangle|^2 |\langle k_n|E\rangle|^2 \\ &\simeq \frac{1}{\mathcal{N}} \sum_E |\langle k_0|E\rangle|^2 \; e^{-n/\xi_{|E\rangle}} \end{aligned} \tag{35}$$

with $\mathcal{N}$ chosen to ensure the normalisation $\sum_n \Lambda_n = 1$ (Eq. 10). The reason we posit a distribution of lengthscales $\xi_{|E\rangle}$ over eigenstates at the level of each realisation

is because, as demonstrated in the preceding section, exact diagonalisation (ED) results (Fig. 7) show that $\Lambda_n$ exhibits stretched exponential behaviour for typical disorder realisations as well as for the disorder-averaged $\langle\Lambda_n\rangle$; so stretched exponential behaviour is not solely due to averaging over disorder realisations. Eq. 35 can be written as

$$\Lambda_n = \frac{1}{\mathcal{N}}\int_0^\infty d\xi\ e^{-n/\xi} P_{\xi_{|E\rangle}}(\xi) \tag{36}$$

with $P_{\xi_{|E\rangle}}(\xi)$ given by

$$P_{\xi_{|E\rangle}}(\xi) \ = \ \sum_E |\langle k_0|E\rangle|^2\ \delta\big(\xi - \xi_{|E\rangle}\big). \tag{37}$$

The form of the lower right hand side of Eq. 35 for $\Lambda_n$ is conceptually apt, as reflected in $|\langle k_0|E\rangle|^2$ appearing explicitly in the summand. So if $|\langle k_0|E\rangle|^2$ for some given eigenstate $|E\rangle$ is 'negligibly small', then that term in the summand makes a negligible contribution to $\Lambda_n$. That is physically sensible, since only a vanishing fraction of eigenstates make a significant contribution to the eigenstate sum in the first line of Eq. 35.[2] Relatedly, that $|\langle k_0|E\rangle|^2$ appears in Eq. 37 reflects that $P_{\xi_{|E\rangle}}(\xi)$ is the distribution of $\xi$ for those eigenstates which make a non-negligible contribution to $\Lambda_n$: the $|\langle k_0|E\rangle|^2$s in effect control the fraction of eigenstates which can make a non-negligible contribution to the eigenstate sum in Eq. 37.

From Eq. 36, the normalisation $\sum_n \Lambda_n = 1$ gives (for $N_{\mathcal{H}} = 2^L \gg 1$) that $\mathcal{N} \equiv \int_0^\infty d\xi\ P_{\xi_{|E\rangle}}(\xi) \int_0^\infty dn\ e^{-n/\xi} = \int_0^\infty d\xi\ \xi\ P_{\xi_{|E\rangle}}(\xi) = \xi_{\rm d}$. The profile of $\Lambda_n$ for a given disorder realisation can thus be expressed as

$$\Lambda_n = \frac{1}{\xi_{\rm d}}\int_0^\infty d\xi\ e^{-n/\xi} P_{\xi_{|E\rangle}}(\xi) \tag{38}$$

which, using the form Eq. 33 for $P_{\xi_{|E\rangle}}(\xi)$, yields the scaling form

$$\Lambda_n = \frac{1}{\xi_{\rm d}} \times g\left(n/\xi_{\rm d}\right) \tag{39}$$

with $g$ a function solely of $n/\xi_{\rm d}$. The disorder-averaged $\langle\Lambda_n\rangle$ is correspondingly given by

$$\langle\Lambda_n\rangle = \int_0^\infty d\xi_{\rm d}\ P_{\xi_{\rm d}}(\xi_{\rm d})\ \Lambda_n, \tag{40}$$

and hence via Eqs. 39,34 is of form

$$\langle\Lambda_n\rangle = \frac{1}{\bar{\xi}} \times f\left(\frac{n}{\bar{\xi}}\right) \tag{41}$$

where $f$ is a function solely of $n/\bar{\xi}$. This is precisely the scaling behaviour found numerically (Eq. 30 with $\beta = \alpha$), viz. $\langle\Lambda_n\rangle = N_{\mathcal{H}}^{-\alpha} f(n/N_{\mathcal{H}}^{\alpha})$. It also identifies the Krylov chain localisation length $\bar{\xi} \propto N_{\mathcal{H}}^{\alpha}$; which, while exponentially large in the system size $L$, is a vanishing fraction of the chain length $N_{\mathcal{H}}$ (as $\alpha < 1$), again symptomatic of the many-body localised nature of the eigenstates.

Although the discussion above is independent of the precise forms of the distributions in Eqs. 33,34, we can infer a specific form for the distribution $P_{\xi_{\rm d}}(\xi_{\rm d})$ of localisation lengths $\xi_{\rm d}$ over disorder realisations. From $S_{K,\infty} = \sum_n n\Lambda_n \equiv \int_0^\infty dn\ n\Lambda_n$, together with Eqs. 38,33, it is easily seen that $S_{K,\infty} = u\xi_{\rm d}$ (with $u$ a constant), and hence the disorder-averaged $\langle S_{K,\infty}\rangle = u\langle\xi_{\rm d}\rangle = u\bar{\xi}$; such that

$$s = \frac{S_{K,\infty}}{\langle S_{K,\infty}\rangle} = \frac{\xi_{\rm d}}{\bar{\xi}} \tag{42}$$

is equivalently just the ratio of $\xi_{\rm d}$ to its mean value $\bar{\xi}$. The distributions $P_s(s)$ and $P_{\xi_{\rm d}}(\xi_{\rm d})$ are thus related by $P_{\xi_{\rm d}}(\xi_{\rm d}) = (1/\bar{\xi})P_s\big(s = \xi_{\rm d}/\bar{\xi}\big)$. The fact that $P_s(s) = e^{-s}$ is found from numerics to be an exponential distribution (see Fig. 3 and Eq. 28) in turn implies that $P_{\xi_{\rm d}}(\xi_{\rm d})$ is also an exponential distribution, specifically

$$P_{\xi_{\rm d}}(\xi_{\rm d}) = \frac{1}{\bar{\xi}}\ \exp\left[-\xi_{\rm d}/\bar{\xi}\right]. \tag{43}$$

Motivated by this, we assume also the simplest such exponential distribution for $P_{\xi_{|E\rangle}}(\xi)$,

$$P_{\xi_{|E\rangle}}(\xi) = \frac{1}{\xi_{\rm d}}\ \exp\left[-\xi/\xi_{\rm d}\right]. \tag{44}$$

With this, Eq. 38 gives explicitly

$$\Lambda_n = \frac{2}{\xi_{\rm d}}\sqrt{\frac{n}{\xi_{\rm d}}}\ K_1\left(2\sqrt{\frac{n}{\xi_{\rm d}}}\right), \tag{45}$$

with $K_1$ a modified Bessel function of the second kind. From the large-$x$ asymptotics $K_1(x) \propto e^{-x}$, Eq. 45 shows that $\Lambda_n$ indeed has a stretched exponential decay, with a stretch exponent of $\gamma = 1/2$. However, we still need to average the profile over disorder realisations as in Eq. 40, and from Eqs. 43,45 this yields

$$\langle\Lambda_n\rangle \ = \ \frac{1}{\bar{\xi}} y^{1/2}\ \mathcal{I}(y) \qquad : \ y = 4n/\bar{\xi} \tag{46}$$

where $\mathcal{I}(y) = 2\int_0^\infty du\ e^{-1/u^2} K_1(\sqrt{y}u)$. While the explicit form of this integral (a Meijer G-function) is not illuminating, its large-$y$ asymptotics are. In particular, it can be shown that

$$\frac{\ln\left[-\ln\mathcal{I}(y)\right]}{\ln y} \sim \frac{1}{3} + \frac{\ln\left[3/2^{2/3}\right]}{\ln y} + \frac{2^{-1/3}}{3} y^{-1/3} + \cdots \tag{47}$$

which in turn implies that the stretch exponent for the stretched exponential decay of $\langle\Lambda_n\rangle$ is in fact $\gamma = 1/3$.

[2] For example, a rough caricture of MBL eigenstates takes the $|\langle k_0|E\rangle|^2$ to be non-zero and of order $\sim N_{\mathcal{H}}^{-D}$ $(D < 1)$ only for $\sim N_{\mathcal{H}}^{D}$ eigenstates (such that $\sum_E |\langle k_0|E\rangle|^2 = 1$); so that only a vanishing fraction $\sim N_{\mathcal{H}}^{D-1}$ of eigenstates contribute to $\Lambda_n$.

As discussed in the preceding section (IV B) with regard to Fig. 7, this is also consistent with the numerical data, particularly at large values of $n/\bar{\xi} \propto n/N_{\mathcal{H}}^{\alpha}$. That being said, the convergence of $\gamma$ to its asymptotic value of $1/3$ is extremely slow, as evident from the subleading corrections in Eq. 47. In fact, the full $\ln[-\ln \mathcal{I}(y)]/\ln y$ throughout the regime $10 \lesssim y \lesssim 100$ – as relevant to the numerical results shown in Fig. 7 –lies within $\sim 0.05$ of $1/2$. This offers a possible rationale for the numerical observation of $\gamma \simeq 1/2$ over the range of practically accessible system sizes, but which ultimately tends asymptotically to $1/3$.

## V. STATISTICS OF EIGENSTATE CONTRIBUTIONS TO SPREAD COMPLEXITY

The infinite-time spread complexity can be written as a sum of contributions from individual many-body eigenstates. Specifically, Eq. 8 may be recast as

$$S_{K,\infty} = \sum_E S_{K,|E\rangle} \,, \tag{48}$$

where

$$S_{K,|E\rangle} = \sum_n n\, \Lambda_{n,|E\rangle} = \sum_n n\, |\langle k_0|E\rangle|^2 |\langle k_n|E\rangle|^2 \tag{49}$$

defines the Krylov complexity associated with the eigenstate $|E\rangle$. This decomposition naturally leads to the question of which eigenstates dominate the sum, and what statistical properties of $S_{K,|E\rangle}$ control $S_{K,\infty}$.

One expects eigenstates with anomalously large $S_{K,|E\rangle}$ to correspond to wavefunctions whose support on the Krylov chain extends to atypically high Krylov orbitals, signalling the presence of long-ranged resonances on the Krylov chain. The central question we address in this section is whether such atypical eigenstates exist and, if so, how their statistics imprint themselves on the infinite-time spread complexity $S_{K,\infty}$; more precisely, whether $S_{K,\infty}$ is built from a finite fraction of eigenstates, or is instead dominated by a vanishing fraction residing in the extreme tails of the distribution of $S_{K,|E\rangle}$.

To first obtain a broad brush view we define

$$w_{|E\rangle} = S_{K,|E\rangle}/S_{K,\mathrm{typ}} \,, \tag{50}$$

where $w_{|E\rangle}$ is the complexity of the eigenstate relative to the typical eigenstate complexity defined via $\ln S_{K,\mathrm{typ}} = N_{\mathcal{H}}^{-1} \langle \sum_E \ln S_{K,|E\rangle}\rangle$. Numerical results for the distribution of $w_{|E\rangle}$ over both eigenstates and disorder realisations, $P_w(w) = N_{\mathcal{H}}^{-1} \langle \sum_E \delta(w - w_{|E\rangle})\rangle$, are shown in Fig. 8. From this it is evident that in the ergodic phase, the distribution is converged with $L$ and well behaved with finite support. By contrast, in the MBL phase the distribution is significantly broader and develops longer tails with increasing system size. This is the first indication that $S_{K,\infty}$ in the MBL phase is indeed dominated by a few eigenstates whose $S_{K,|E\rangle}$ lie in the tails of $P_w$.

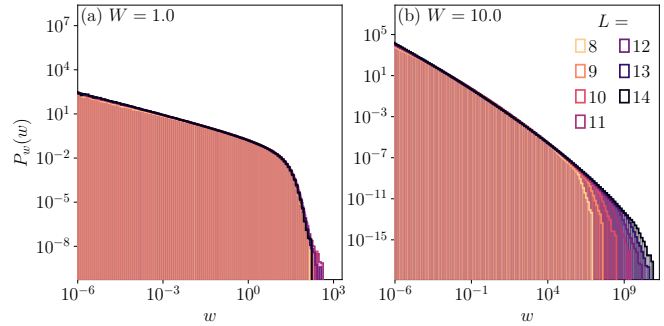


FIG. 8. Probability distribution $P_w(w)$ of $w_{|E\rangle} = S_{K,|E\rangle}/S_{K,\mathrm{typ}}$, over eigenstates $|E\rangle$ and disorder realisations. Panel (a) shows results for the ergodic phase at $W = 1$, and panel (b) for the MBL phase at $W = 10$. The data shows the development of increasingly longer tails in $P_w(w)$ with increasing $L$ in the MBL phase, but not in the ergodic phase. This is highlighted by the remarkably large difference in the scales of the support of $P_w(w)$.

To characterise this quantitatively, we introduce the generalised moments

$$S_{K,\infty}^{(q)} = \sum_E S_{K,|E\rangle}^q \,, \tag{51}$$

which allow for a large-deviation analysis analogous to that on the Fock-space graph [71, 74]. Defining $z_{|E\rangle} = \ln S_{K,|E\rangle}$, the disorder-averaged moments can be written as

$$\langle S_{K,\infty}^{(q)} \rangle = \int dz\; P_z(z)\, e^{qz}\, e^{L \ln 2} \,, \tag{52}$$

where $P_z(z) = N_{\mathcal{H}}^{-1} \langle \sum_E \delta(z - z_{|E\rangle})\rangle$ is the distribution of $z_{|E\rangle}$ over eigenstates and disorder realisations.

$S_{K,|E\rangle}$ is typically expected to scale polynomially with $N_{\mathcal{H}}$, and thus $z_{|E\rangle}$ to scale extensively with $L$. It is therefore convenient to introduce the intensive variable $x = z/L$, in terms of which Eq. 52 becomes

$$\langle S_{K,\infty}^{(q)} \rangle = \int dx\; P_x(x)\, e^{L \ln 2}\, e^{Lqx} \,, \tag{53}$$

where $P_x(x) = L P_z(Lx)$. The quantity

$$N_x = P_x(x) e^{L \ln 2} \tag{54}$$

represents the total number density of eigenstates contributing a weight $e^{Lqx}$ to $S_{K,\infty}^{(q)}$. This naturally defines a corresponding entropy density

$$\Sigma(x) = \frac{1}{L} \ln N_x = \ln 2 + \frac{1}{L} \ln P_x(x) \,, \tag{55}$$

such that

$$\langle S_{K,\infty}^{(q)} \rangle = \int dx\; \exp\left( L\left[\Sigma(x) + qx\right] \right) . \tag{56}$$